

★★ RECONCILED (K815 + K829 — read FIRST; resolves Elie’s corpus-inconsistency flag). The body table/summary rows “hypercharge = SO(2)=J” are RETRACTED. Correct current view of the hypercharge origin: it is the Z_6 CENTER (K806, my charge row), NOT the SO(2)=J (that U(1) is Cartan-like $\sim T^3_L$, chirality-blind — retracted K815). And the Z_6 center is geometric on both legs (K829): $Z_6 = Z_{\{N_c\}}$ (color center — my $1/N_c$ fractionalization, T2521, target-innocent) $\times$ Z_{rank} (isospin center — the *center* of the U(2) Kähler holonomy, K813). So K813 (the U(2) holonomy is real) and K806 (hypercharge = the Z_6 center) are consistent — hypercharge comes from the U(2)’s *center*, not the SO(2)=J Cartan. This is the reconciliation Elie asked for, and it *strengthens* the charge sector: the isospin term of the center correlation ($6Y \equiv 4t + 3d$) is grounded in the U(2) holonomy. Non-circular, resting on one already-banked target-innocent fact (T2521’s color term). The K815 section below has the full retraction; this banner is the reconciled bottom line.

The electroweak structure from D_{IV}^5 — consolidation + the U(2) cross-link (paper-facing) [hypercharge=SO(2) rows RETRACTED — see banner]

Grace | 2026-07-22e | consolidation pull: the weak sector is now a coherent $\{n_C \text{ odd} + D_{IV}^5 \text{ Hermitian}\}$ result. My lane: consolidate/tier it for the paper, and — the genuinely-mine piece — verify the cross-link, because the electroweak U(2) here is the SAME U(2) whose Z_6 center gave the 1/6 hypercharge quantization in my charge row (K806). Two separate threads landing on one object. Consistency win, derived-conditional, nothing banks.

The consolidated synthesis (K813) — $\{n_C \text{ odd} + \text{Hermitian}\}$, one input

The whole electroweak chiral+selection structure = the isotropy $K = \text{SO}(5) \times \text{SO}(2)$: | SM fact | geometric source | tier | |—|—|—| | chiral SPLIT (spinor $\rightarrow$ L, R halves) | SO(5) [$n_C=5$ odd] $\rightarrow$ spinor 4 irreducible $\rightarrow$ SO(4) split $(2,1) \oplus (1,2)$ | derived | | which half is L (parity orientation) $\rightarrow$ maximal | SO(2)=J [Kähler] $\rightarrow$ canonical orientation picks self-dual = SU(2)_L | derived (closes K809 gap) | | **weak = SU(2)_L, not SU(2)_R** | J-charges: SU(2)_L $\{0, \pm 2\}$ charged, SU(2)_R $\{0, 0, 0\}$ neutral $\rightarrow$ only L can mix with U(1) | derived (Lyra J-computation) | | **hypercharge U(1)_Y** | = SO(2)=J itself | derived (corrects K812: $Y=J$, not “broken SU(2)_R”) | | no W_R (SU(2)_R ungauged) | Kähler reduces SO(4) holonomy $\rightarrow$ U(2); the anti-self-dual factor is cut to U(1) | derived (FA-positive) | | custodial SU(2) / $\rho \approx 1$ | O = (2,2) $\rightarrow$ diagonal SU(2)_V | derived (0.04%, FA bonus) | | CP free | SO(2)=J continuous S^1 phase $\rightarrow$ δ unfixed | banked (K791) | | CP small (J_{CKM}) | rank-1 angles (separate from the phase) | banked (T2519) | One input: the gravi-weak “promote a chiral half to a gauge field” move (web-grounded arXiv:1212.5246, FA-safe, cross-links BST’s SO(5,2) gravity). Not a GUT.

★ The U(2) cross-link (my contribution — two threads, one object)

The electroweak group is $U(2) = \text{SU}(2)_L \times \text{U}(1)_Y$ = the Kähler holonomy** of D_{IV}^5 (K813). And that U(2) is the same one whose center quotient gave me the hypercharge quantization in the charge row (K806):

parity/chirality thread (K813): Kähler holonomy $\rightarrow \text{SU}(2)_L \times \text{U}(1)_Y = \text{U}(2)$
charge-quantization thread (K806): hypercharge quantum = $1/(N_c \cdot \text{rank}) = 1/6$
from $Z_{\{N_c\}} \times Z_{\text{rank}} = Z_3 \times Z_2 = Z_6$ center
THE LINK: the “rank” in my $1/(N_c \cdot \text{rank})$ = the SU(2)_L doublet-center $Z_{\text{rank}} = Z_2$
which the weak-sector thread INDEPENDENTLY derives (Kähler self-dual holonomy).
color center $Z_{\{N_c\}} \times \text{SU}(2)_L$ center $Z_{\text{rank}} = Z_6 \rightarrow 1/6$ hypercharge.

So the two threads aren’t two claims — they’re one U(2) seen twice. The rank I used for the hypercharge quantum is the SU(2)_L the weak sector derives; the color $\times$ doublet centers that give 1/6 are the centers of that same U(2). This is the kind of consistency you want — independent threads (charge quantization, parity/chirality) meeting on one geometric object — not a new derivation, a coherence check that passes.

❓ Discipline (held)

- Weinberg: the electroweak *structure* is derived (only SU(2)_L mixes with the U(1), from the J-charges) — but $\sin^2\theta_W$ is a running, scheme-dependent VALUE → Tier-2, DON'T chase. And the Five-Absence trap: $\sin^2\theta_W = 3/8$ is the GUT value — forbidden. Bank the structure, hold the value.
- Tier: everything derived-conditional — reasons-for-the-SM, not new physics. Rides one gravi-weak input + Keeper's two K812 gaps (U(2) holonomy of internal directions must align with spacetime spinor chirality; self-dual↔left must survive on the curved domain — both handed to Lyra). Nothing banks until those close.
- Five-Absence-positive: U(2) is the electroweak group, not a GUT; it *forbids* W_R rather than adding gauge bosons. No unification, no proton decay.
- The U(2) cross-link is a consistency win, NOT a new bank — it rides the same derived-conditional inputs; I flag it as coherence, not a result.

Net (paper-facing consolidation)

- The weak sector is a coherent {n_C odd + D_{IV}⁵ Hermitian} result: chirality (SO(5) split) + parity-maximal + weak=SU(2)_L + hypercharge=J + no-W_R + custodial/ $\rho \approx 1$ + CP-free — all from two defining features of the domain plus one gravi-weak input. “Why is the world left-handed / why hypercharge / why no W_R / why $\rho \approx 1$ ” → “because n_C is odd and the domain is complex.”
- ★ Cross-link (mine): the electroweak U(2) = the same U(2) whose Z₆ center quantized hypercharge in units of 1/6 (K806) — two threads, one object, a passing coherence check.
- Discipline held: structure derived, Weinberg *value* Tier-2 (running + GUT-trap 3/8); derived-conditional on one gravi-weak input + Keeper's two gaps; FA-positive; nothing banks.

★★ K815 RETRACTION — this consolidation OVER-CONSOLIDATED on a synthesis that didn't hold

Lyra caught it by computing: the SO(2) commutes with SO(5) → chirality-blind → it does NOT orient parity and it is NOT hypercharge. Elie's $J = L_{12} + L_{34}$ is an *internal* SO(4) generator (the arithmetic is right, the *identification* with the domain SO(2) is wrong). So the “one SO(2)=J does three jobs” synthesis I consolidated is partially retracted. Correcting my own table: | claim (above) | K815 status | |—|—| | chiral SPLIT (SO(5), n_C odd) | SURVIVES — solid rep theory | | parity orientation via SO(2)=J → maximal | RETRACTED — SO(2) chirality-blind; orientation now OPEN (candidate: $g=7\omega$, single manifold) | | weak = SU(2)_L via *domain* J-charges | RETRACTED — the J computed was internal SO(4), not the domain SO(2) | | hypercharge = SO(2)=J | RETRACTED — that U(1) is Cartan-like ($\sim T^3_L$), not Y; hypercharge origin OPEN | | no-W_R, custodial $\rho \approx 1$ (O=(2,2)) | SURVIVES — clean, Five-Absence-positive | | CP free / CP small | SURVIVES (K791 / T2519) |

★ My U(2) cross-link — correction: I claimed the EW U(2) = Kähler holonomy = the same U(2) whose Z₆ center gave my 1/6 hypercharge. That link rode the retracted “SO(2)=J=Y” identification. What SURVIVES independently: the charge-row $1/6 = 1/(N_c \cdot \text{rank})$ from $Z_{\{N_c\}} \times Z_{\text{rank}} = Z_6$ (K806) — and Lyra confirms *this* is the clean hypercharge handle, NOT the SO(2)=J. So the honest state: the 1/6-from-the-center handle stands (K806, self-contained); the “it's the Kähler-holonomy U(2) that also gives parity” bridge is retracted with the synthesis. Two threads did NOT provably meet — I over-read a coherence that depended on the conflated identification. The charge-row handle is the surviving, correct one.

What to bank (smaller, honest): the chiral split, CP-free, custodial/ $\rho \approx 1$ /no-W_R — plus the charge-row hypercharge handle (1/6 from the center, independent). OPEN/candidate: parity orientation, hypercharge *origin* on the manifold, the internal→spacetime bridge. The paper scopes DOWN to the survivors; parity/hypercharge stay open. Nothing false banked — the arithmetic caught the overshoot, including in my own consolidation.

— Grace, 2026-07-22e (K815-corrected: survivors banked, parity+hypercharge+cross-link retracted to open). Electroweak consolidation (paper-facing): the whole EW chiral+selection structure = {n_C odd [SO(5) chiral split] + D_{IV}⁵ Hermitian [SO(2)=J: orientation→parity maximal, phase→CP free, =U(1)_Y]}, one gravi-weak

input; weak= $SU(2)_L$ (J-charges $\{0, \pm 2\}$ charged vs $SU(2)_R \{0,0,0\}$ neutral $\rightarrow$ only L mixes), no- W_R (Kähler cuts anti-self-dual to $U(1)$), custodial $\rho \approx 1$ ($O=(2,2)$). ★ MY CROSS-LINK: the EW $U(2)=SU(2)_L \times U(1)_Y$ *Kähler holonomy IS the same $U(2)$ whose $Z\{N_c\} \times Z_{\text{rank}}=Z_6$ center gave my $1/6$ hypercharge quantum (K806) — the ‘rank’ in $1/(N_c \cdot \text{rank})$ = the $SU(2)_L$ center the weak sector derives; two threads one object, coherence check passes. DISCIPLINE: structure derived, $\sin^2 \theta_W$ VALUE Tier-2 running + GUT-trap $3/8$ don’t chase; derived-conditional (1 gravi-weak input + Keeper’s 2 gaps); FA-positive; nothing banks.*